Sinong (Simon) Zhan

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simon-zhan.com | GitHub | Google Scholar

Education

Northwestern University — **Ph.D. expected Dec 2027**
Ph.D., Electrical and Computer Engineering — Advisor: Qi Zhu
Master's, Computer Engineering — Jun 2026
University of California, Berkeley — Dec 2022
B.A., Computer Science and Applied Mathematics — Research advisor: Sanjit A. Seshia

Experience

Google — Jun 2026 – Present
Research Intern — *Google Research & P+D Team*

- Built RL infrastructure on XManager for long-horizon coding agents, supporting **0.5B–30B parameter models** with tensor/sequence parallelism, ZeRO-1 optimizer sharding, and memory management for rollout and training.
- Designed memory-aware RL for coding agents to retain informative context from their own trajectory histories, with separate credit assignment for summarization, reasoning, and tool-call tokens during long-horizon training.

Amazon — Jun 2025 – Sep 2025
Applied Scientist Intern — *Store Foundation AI*

- Developed multi-turn conversational agents for the **Rufus Shopping Assistant** using reinforcement learning; designed logic-based reward machines for human shopping-behavior simulation and LLM-agent training pipelines.
- Contributed to customer-facing systems and LLM-agent training using **AWS EC2/EKS, S3, and CloudWatch**.

Northwestern University, IDEAS Lab — Sep 2023 – Present
Graduate Research Assistant — *Advisor: Qi Zhu*

- Co-built **SENTINEL**, a temporal-logic framework for safety evaluation of foundation-model embodied agents across interpretation, task planning, and execution in VirtualHome and AI2-THOR (**NeurIPS 2026, accepted**).
- Co-developed **ManiGuard**: 200 manipulation tasks, 1,000 scenarios, and 8,000 safety-annotated demonstrations; evaluated task safety with runtime monitors under distribution shift, in simulation and on a physical Franka robot.
- Developed **DT-CORL**, jointly learning transformer-based beliefs and offline policies from fixed delay-free datasets for delay-robust policy deployment (**ICLR 2026**); advanced variational delayed RL (**NeurIPS 2024, Spotlight**).
- Developed barrier-certificate learning for state-wise safe RL from pixels, alongside transition-aware reward shaping for robust and sample-efficient inverse RL under stochastic dynamics and model uncertainty (**L4DC 2024/2026**).

University of California, Berkeley — Feb 2022 – Apr 2023
Undergraduate Researcher & Research Assistant — *Advisor: Sanjit A. Seshia*

- Built verification and control-synthesis workflows for underwater robotics, including path and trajectory planning; created an SMT-based 3D bin-packing solver for automated design model checking in the NSF UUV Challenge.

Selected Publications

*First/co-first-author work; * equal contribution.* — Full publication list and papers

- **S. S. Zhan***, P. Wang*, J. Liu*, Y. Peng*, Y. Lyu, Z. Wang, Q. Wang, et al. **SENTINEL**: A Multi-Level Formal Framework for Safety Evaluation of Foundation Model-based Embodied Agents. *NeurIPS 2026, accepted.* [Paper]
- **S. S. Zhan*** et al. Belief-Based Offline Reinforcement Learning for Delay-Robust Policy Optimization. *ICLR 2026.*
- Q. Wu*, **S. S. Zhan***, Y. Wang, et al. Variational Delayed Policy Optimization. *NeurIPS 2024,* ***Spotlight****.*

Technical Skills

Programming / ML: Python, C/C++, CUDA, SQL; PyTorch, JAX, Hugging Face, verl; reinforcement learning.
Infrastructure: Distributed training, tensor/sequence parallelism, ZeRO-1, XManager, AWS, Docker, Linux, Git.
Robotics / Verification: MuJoCo, Isaac Sim, AI2-THOR, VirtualHome; temporal logic, SMT, Gurobi, certificates.